

Kevin Pulikkottil

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PROFESSIONAL SUMMARY

Computer Science graduate with hands-on experience in embedded security, cryptographic protocol implementation, and software development. Implemented cryptographic key-exchange components for a Toyota TMNA capstone and contributed a merged fix to a 55,000+ star open source project. CompTIA Security+ certified; pursuing CDSA. Seeking entry-level roles in cybersecurity engineering or software engineering.

EDUCATION

The University of Texas at Dallas, Richardson, TX

Bachelor of Science, Computer Science, May 2026

Relevant Coursework: Database Systems, Computer Networks, Networking Laboratory, Operating Systems, Data Structures & Algorithm Analysis, Software Engineering, Digital Logic & Computer Design

EXPERIENCE

Vehicle Cybersecurity Engineer, Embedded Security Project

Toyota Secure Vehicle Communications, UDesign Capstone Project

January 2026 - May 2026

- Built an **end-to-end, X448-based, rekey-capable EV charging handshake** in the **OP-TEE Trusted Application** and host client, meeting Toyota TMNA Product Cybersecurity standards for the **ISO 15118-20** use case.
- Implemented **X448 Elliptic Curve Diffie-Hellman (RFC 8418)** for ephemeral session keys and stabilized **X.509 certificates** across sessions by refactoring the custom CMAC extension out of the TA, builder, parser, and verifier.
- Replaced the team's PyQt6 desktop demo with a browser-based dashboard for the UDesign EXPO, visualizing **live MITM attack detection** between EV, EVSE, and a Raspberry Pi attacker node for corporate stakeholders.

TECHNICAL SKILLS

Programming Languages: Python, C/C++, Java, SQL

Software Engineering: Object-Oriented Design, Data Structures, Algorithms, Software Design Patterns, UNIX Systems

Programming, Database Design and Normalization

Tools & Platforms: Git, Linux, Arm64 QEMU, Raspberry Pi, Tailscale, Pi-hole, scikit-learn, pandas, NumPy, Pytest

Security: Network Security, Firewalls, Firewall Management, DNS Configuration, Network Protocols (OSI Model, TCP/IP), Access Control, Data Integrity, Cryptographic Protocol Implementation, Embedded Security

PROJECTS

- **MemPalace Open Source Contributor (GitHub)**
 - Contributed a merged fix to MemPalace (55,000+ GitHub stars) resolving a user-reported CLI bug, adding a first-class mempalace mcp subcommand that prints copy-pastable MCP server setup guidance with custom palace-path handling, backed by Pytest regression tests.

Secure Home Lab on Raspberry Pi, Personal Project

December 2025 - Present

- Reduced network ad traffic by **99%** by deploying a Pi-hole DNS sinkhole reachable remotely over a Tailscale tailnet.
- Implemented custom firewall rules, strict access controls, and active DNS query log monitoring.

Breast Cancer Detection Model, Algoverse AI Research

February 2024 - May 2024

- Achieved 95.6% diagnostic accuracy by building and tuning a scikit-learn classifier on the Breast Cancer Wisconsin dataset, using pandas and NumPy for data preprocessing.

CERTIFICATIONS

- CompTIA Security+, CompTIA (April 2026)
- AWS Certified Solutions Architect - Associate (SAA-C03), AWS (In Progress)

AWARDS & HONORS

Eagle Scout, Scouting America (January 2021)